Final Report - Enhanced SMS Fraud Detection Using Machine Learning

Abstract

This project extends the work of Batoul Abo Yehya and Nazih Salhab's paper "Telecommunications Fraud Machine Learning-based Detection" (2023), focusing specifically on SMS fraud detection. We replicated their baseline approach and introduced significant enhancements to address modern fraud patterns. Our enhanced SVM model achieves 99.37% accuracy with 97.04% recall, representing a 1.89% improvement in recall over the baseline while maintaining high precision.

Key Contributions: - Dual Word2Vec models for semantic understanding of text and URL patterns - POS-aware preprocessing with context-sensitive lemmatization - 14 engineered features capturing fraud-specific patterns - Combined dataset approach balancing URL-based and social engineering fraud - Comprehensive evaluation demonstrating the value of engineered features

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1. Project Overview

1.1 Original Paper's Approach

The original paper by Abo Yehya and Salhab (2023) addressed two types of telecommunications fraud:

1.1.1 IRSF (International Revenue Share Fraud) Detection

IRSF is a scheme where fraudsters generate artificial traffic to premium-rate numbers, earning revenue share from telecom operators. The paper compared five ML classifiers using Call Detail Records (CDRs):

- Models tested: Random Forest, Logistic Regression, SVM, Gradient Boosting, Naive Bayes
- Features: Call duration, destination country, call frequency, time patterns
- Winner: Random Forest with 99% accuracy
- Key insight: CDR metadata patterns effectively identify IRSF

1.1.2 SMS Fraud Detection

For SMS fraud, the paper used text-based classification:

- Models: SVM and Logistic Regression with TF-IDF vectorization
- Winner: SVM with 99.03% accuracy
- Dataset: 12,315 fraud messages + 40,536 normal messages
- Critical limitation: 94.76% of fraud messages contained URLs

Our Focus: This project extends the SMS fraud detection component, addressing its limitations for modern fraud patterns.

1.2 The SMS Fraud Problem

SMS fraud represents a significant portion of the $39.89 billion in annual telecommunications fraud losses globally. The problem has evolved beyond simple URL-based phishing to include sophisticated social engineering attacks that exploit psychological manipulation rather than technical vulnerabilities.

An overview on two primary categories of Modern SMS fraud:

1. URL-Based Fraud (Traditional)
    - Contains malicious links to phishing sites
    - Easily identifiable by URL presence
2. Social Engineering Fraud (Emerging)
    - No URLs, relies on psychological manipulation
    - Uses urgency, authority, and financial incentives

The shift toward social engineering fraud necessitates more sophisticated detection methods that understand context and semantic meaning, not just keyword matching.

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2. Methodology

2.1 Limitations of the Original SMS Approach

While the original paper achieved high accuracy, several limitations emerged:

1. Dataset Bias - 94.76% of fraud messages contained URLs - Models could achieve high accuracy by simply detecting URLs - Doesn't test ability to detect social engineering fraud without URLs

2. Feature Limitations - TF-IDF only: Cannot capture semantic relationships - No context understanding: "urgent" and "immediate" treated as unrelated - Missing fraud-specific patterns: No features for urgency, financial terms, or text statistics

3. Generalization Concerns - Modern fraud increasingly uses social engineering without URLs - Model may fail on sophisticated attacks that avoid obvious keywords - No evaluation on diverse fraud types

The original dataset's 94.76% URL presence means models never learned to detect these sophisticated attacks, creating a critical blind spot for modern fraud detection.

Figure 1 shows an exaggerated version of social engineered fraud message, illustrating how modern fraud relies on psychological manipulation (urgency, authority, financial incentives) rather than URLs.

2.2 Our Dataset Solution: Balancing URL-Based and Social Engineering Fraud

To address the URL bias, we created a balanced dataset combining two sources:

Dataset 1: Original Telecom Fraud Dataset - Source: nsalhab/TelecomFraudDetection GitHub repository - 12,315 fraud messages (primarily URL-based) - 40,536 normal messages

Dataset 2: UCI SMS Spam Collection - Source: UCI Machine Learning Repository - Citation: Almeida, T.A., Gómez Hidalgo, J.M., Yamakami, A. (2011) - 747 spam messages (social engineering focus) - 4,827 legitimate messages

Balancing Strategy: We downsampled the original dataset to match the UCI size and created a 50:50 ratio of URL-based versus social engineering fraud, resulting in a final combined dataset of approximately 10,000 messages. This balanced approach ensures that models learn both fraud patterns equally, preventing bias toward URL-based detection.

This balanced approach tests whether our enhanced features can generalize beyond simple URL detection to understand contextual fraud indicators.

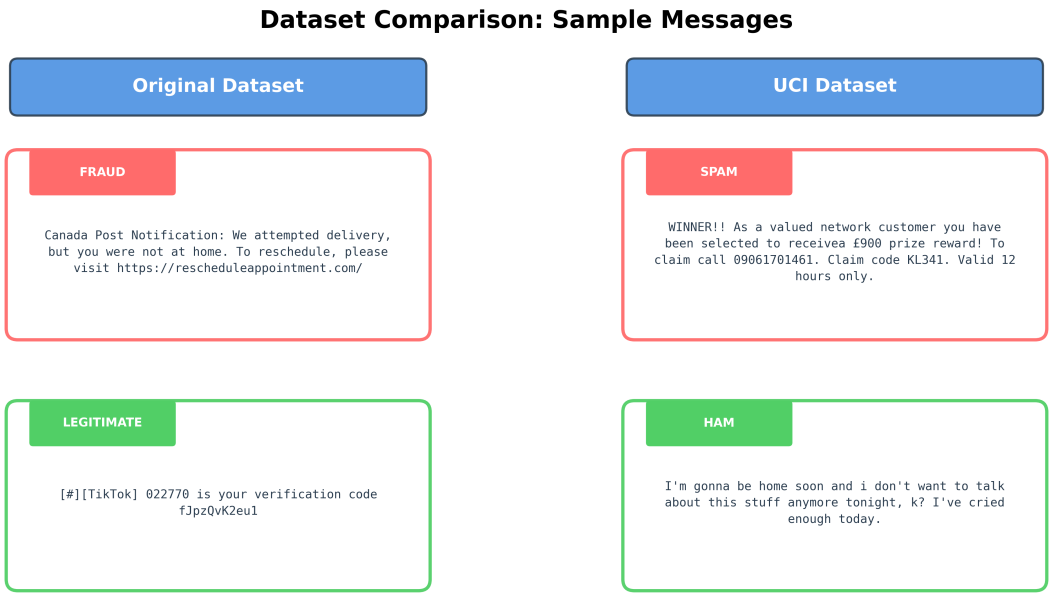

Figure 2 compares the difference between the original and UCI datasets, highlighting the URL bias problem and our balanced solution.

2.3 TF-IDF: The Baseline Approach

TF-IDF (Term Frequency - Inverse Document Frequency) identifies important words by balancing term frequency with document rarity.

Example:

Consider this fraud message: *"URGENT! You won $1000! Click here NOW!"*

- High TF-IDF scores: "urgent", "won", "click" (fraud-specific words)
- Low TF-IDF scores: "you", "here" (common everywhere)

TF-IDF successfully identifies fraud keywords, making it effective for URL-based fraud detection.

Limitations:

While TF-IDF works well for keyword matching, it has critical weaknesses:

- No context understanding: "bank" (river) vs "bank" (money) → same score
- Ignores word order: "dog bites man" = "man bites dog"
- No semantic relationships: Doesn't know "urgent" "immediate" "hurry"
- Misses manipulation patterns: Can't detect psychological tactics without specific keywords

These limitations are problematic for social engineering fraud, which relies on contextual manipulation rather than specific keywords.

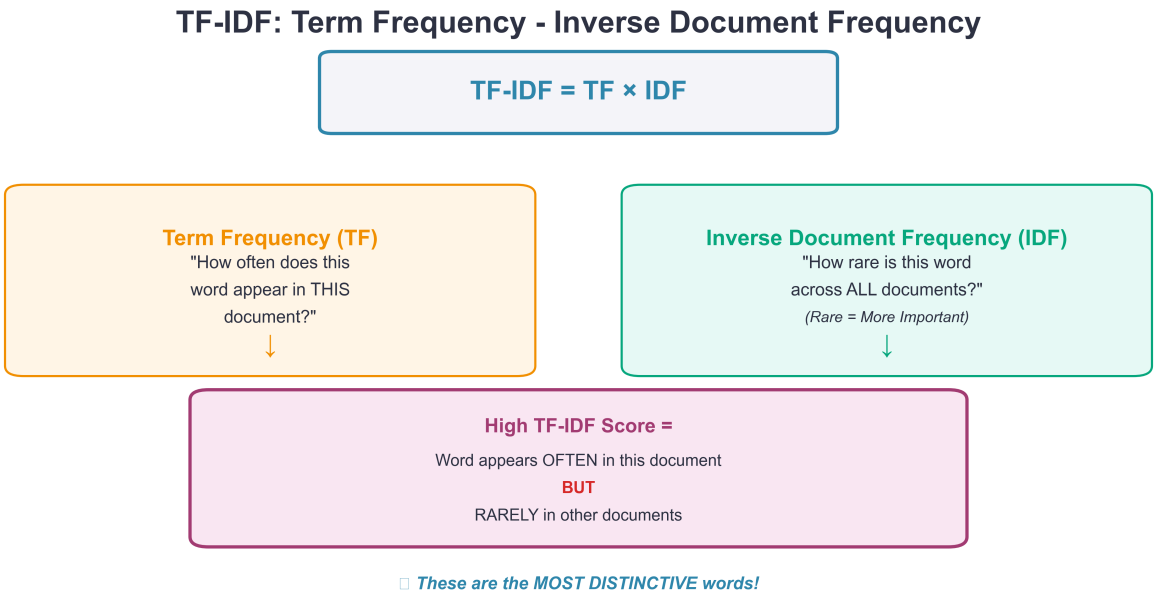

Figure 3 illustrates the TF-IDF design, showing how it balances term frequency with document rarity to identify important words.

2.4 Word2Vec: Understanding Context

To address TF-IDF's limitations, we introduced **Word2Vec** embeddings that capture semantic relationships between words.

Word2Vec is built on the distributional hypothesis, elegantly captured by J.R. Firth's (1957) principle: "You shall know a word by the company it keeps." This foundational idea suggests that words appearing in similar contexts tend to have similar meanings. Through this training process, it creates dense vector representations — in this case, 100 dimensions - that capture semantic similarity in a continuous space rather than discrete keyword matching.

Consider two fraud messages: *"URGENT! Claim your prize now!"* and *"Act IMMEDIATELY to receive your reward!"* While TF-IDF treats "urgent" and "immediately" as completely different words, Word2Vec learns they mean similar things by observing their similar contexts—both appear before action words like "claim" and "act." Similarly, it learns "prize" "reward" because both appear in winning contexts alongside words like "win," "receive," and "congratulations." This semantic understanding allows Word2Vec to detect both messages as fraud despite different vocabulary, addressing TF-IDF's fundamental limitation of treating synonyms as unrelated terms. The model generalizes beyond exact keyword matches to understand the underlying intent and manipulation tactics.

Technically, Word2Vec converts each word into a one-hot encoded vector, then trains a shallow neural network to predict context word probabilities. During training, the network adjusts its weights to maximize the probability of predicting correct context words. The hidden layer's weights become the word embeddings: dense vectors that encode semantic meaning learned from predicting word co-occurrences across the training context. These embeddings are far more compact than one-hot vectors while capturing rich semantic information.

The resulting embedding space has a remarkable property: semantically related words cluster together, and their relationships are preserved through vector arithmetic. Words with similar meanings (like "urgent" and "immediate") have embeddings that are close in this high-dimensional space, measurable by cosine similarity—a metric. This geometric structure enables the model to recognize fraud patterns even when dif-

ferent vocabulary is used, as semantically equivalent phrases map to nearby regions in the embedding space. Famous examples include vector("king") - vector("man") + vector("woman") vector("queen"), demonstrating how semantic relationships become mathematical operations in this learned space.

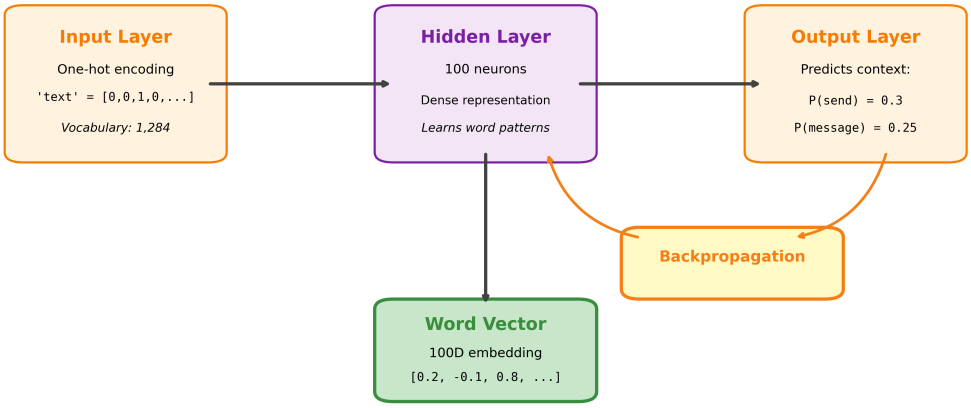

Figure 4 demonstrates how Word2Vec is implemented, using a neural network to learn word relationships from context.

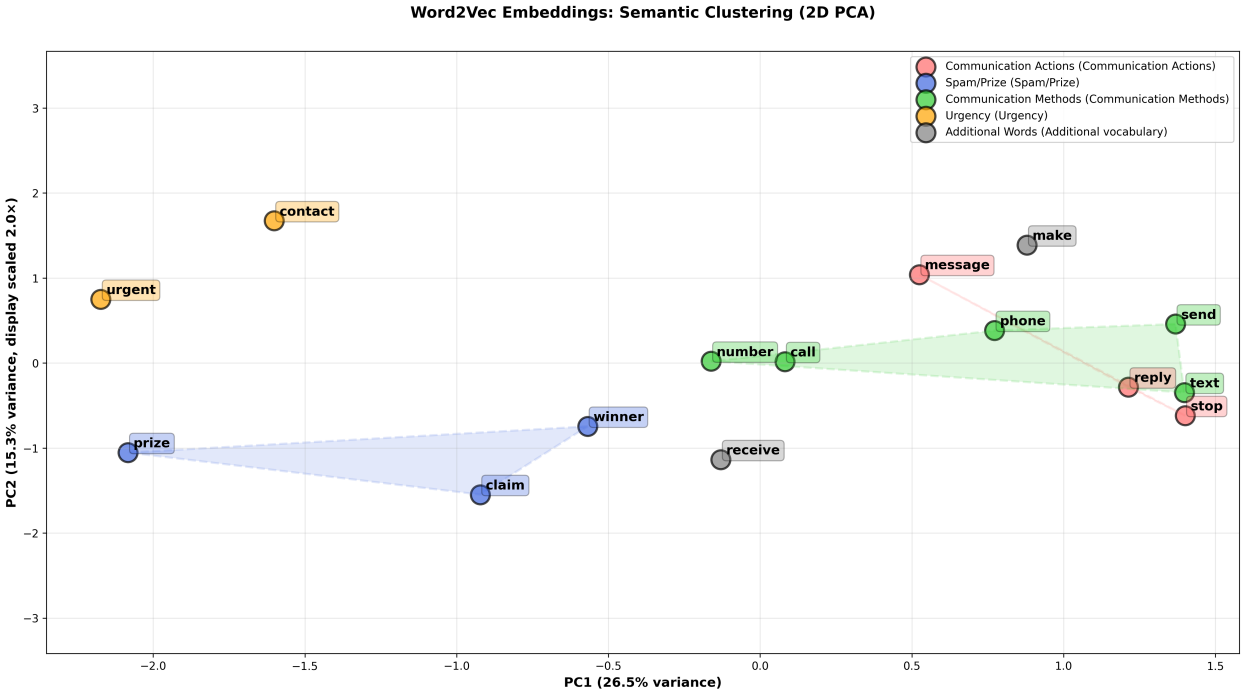

Figure 5 picked out some word clusters from trained Word2Vec using the UCI dataset, demonstrating how semantically related words cluster together in vector space.

Our implementation employs a dual Word2Vec approach to capture both textual and URL-based fraud

patterns. The first model, Text Word2Vec (100 dimensions), is trained on cleaned message text to capture semantic relationships in natural language, learning how fraudsters use urgency, financial incentives, and psychological manipulation. The second model, URL Word2Vec (100 dimensions), is trained separately on URL patterns to capture malicious domain structures, identifying suspicious characteristics like shortened URLs, misspelled domains, and unusual TLDs that often indicate phishing attempts. This dual approach ensures comprehensive feature extraction from both message content and embedded links.

2.5 POS-Aware Preprocessing

Part-of-Speech (POS) tagging enables context-sensitive lemmatization:

- "You **won** the prize" (verb) → keep as "won"
- "A big **win** for you" (noun) → keep as "win"
- "The **winning** ticket" (adjective) → lemmatize to "win"

Pipeline Steps:

1. Tokenization: Split text into words
2. POS Tagging: Identify grammatical role (noun, verb, adjective, etc.)
3. Context-Sensitive Lemmatization: Convert to base form based on POS tag
4. Stop Word Removal: Remove common words ("the", "a", "is")
5. URL Extraction: Separate URLs for dedicated Word2Vec processing

This preprocessing ensures that our Word2Vec models learn from clean, semantically consistent and properly normalized text, improving the quality of learned embeddings.

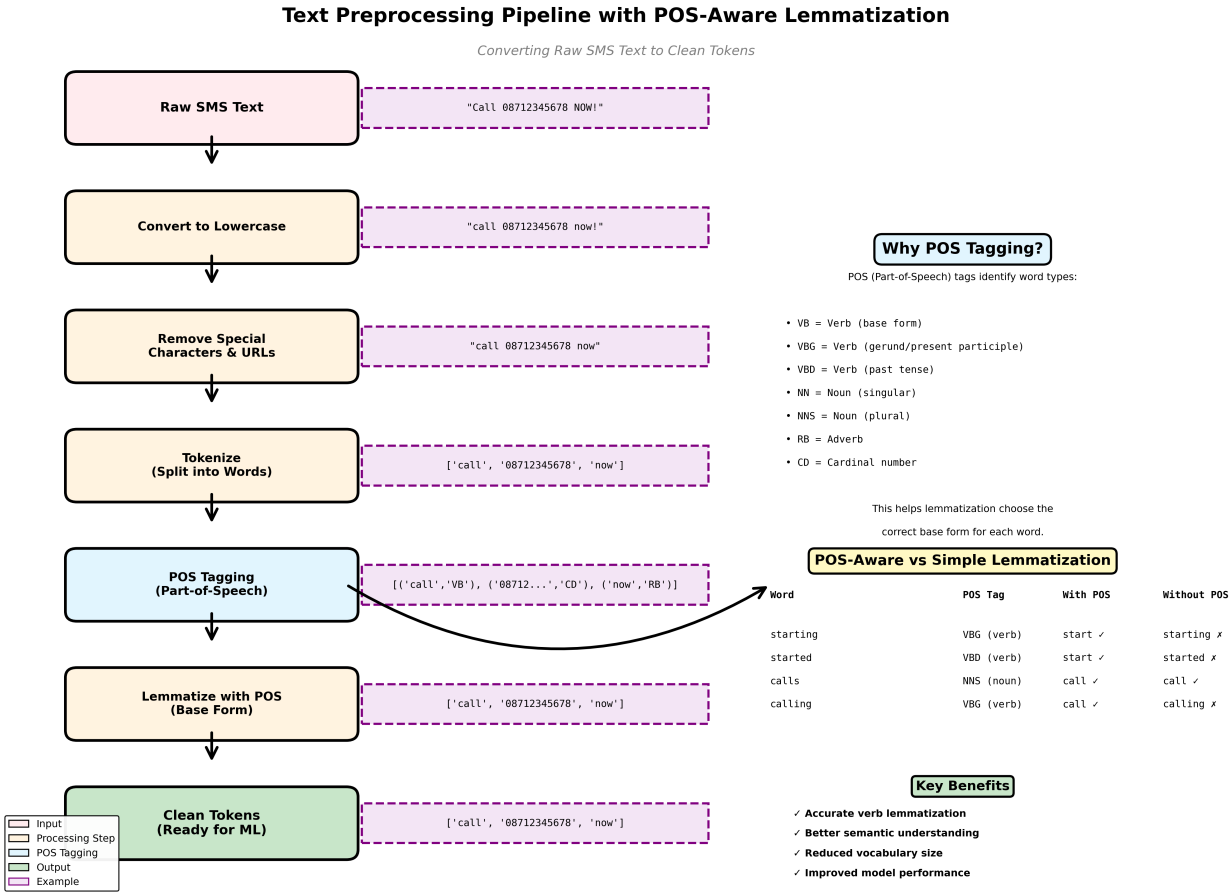

Figure 7 shows an overall summary of the enhanced data preprocessing pipeline, illustrating our five-step

approach to text cleaning.

2.6 Fourteen Engineered Features

Beyond TF-IDF and Word2Vec, we manually crafted 14 features capturing fraud-specific patterns that semantic models might miss. These features encode domain knowledge about fraud patterns—for example, fraudsters often use urgency ("act now!"), financial incentives ("win $1000"), and excessive punctuation ("!!!") to manipulate victims.

14 Engineered Features for SMS Fraud Detection

Category	Feature Name	Description	Fraud Indicator
Text Statistics	**text_length**	Total character count	Longer messages build credibility
Text Statistics	**word_count**	Number of words	More words = more persuasion
Special Characters	**exclamation_count**	Number of '!' marks	Creates urgency/excitement
Special Characters	**question_count**	Number of '?' marks	Engagement tactics
Special Characters	**dollar_count**	Number of '$' signs	Financial focus
Special Characters	**percent_count**	Number of '%' signs	Discount/offer indicators
Text Composition	**uppercase_ratio**	Ratio of UPPERCASE letters	SHOUTING = spam tactic
Text Composition	**digit_ratio**	Ratio of digits (0-9)	Phone numbers, codes, amounts
URL Analysis	**has_url**	Boolean: contains URL	69% fraud vs 2% legitimate
URL Analysis	**url_count**	Number of URLs	Multiple URLs = suspicious
URL Analysis	**avg_url_length**	Average URL length	Shortened URLs (bit.ly) = fraud
URL Analysis	**has_https**	Boolean: has HTTPS	Legitimate sites use secure links
Behavioral	**urgency_score**	Count of urgency words	'urgent', 'immediately', 'expire'
Behavioral	**financial_score**	Count of financial words	'bank', 'account', 'payment'

Figure 6 explains the 14 engineered features across four categories: URL analysis, urgency indicators, financial indicators, and text statistics.

These 14 features complement TF-IDF and Word2Vec by capturing structural and stylistic fraud indicators that pure text analysis might miss.

Why These Features Matter:

Fraud messages exhibit distinctive patterns beyond word choice:

- Urgency: Create pressure for immediate action ("now", "urgent", "immediately")
- Financial incentives: Promise money, prizes, rewards
- ALL CAPS: Grab attention, appear authoritative
- Excessive punctuation: "!!!" creates urgency and excitement
- URLs: Direct victims to phishing sites
- Short, punchy messages: Quick, impactful communication

2.7 Model Architecture and Training

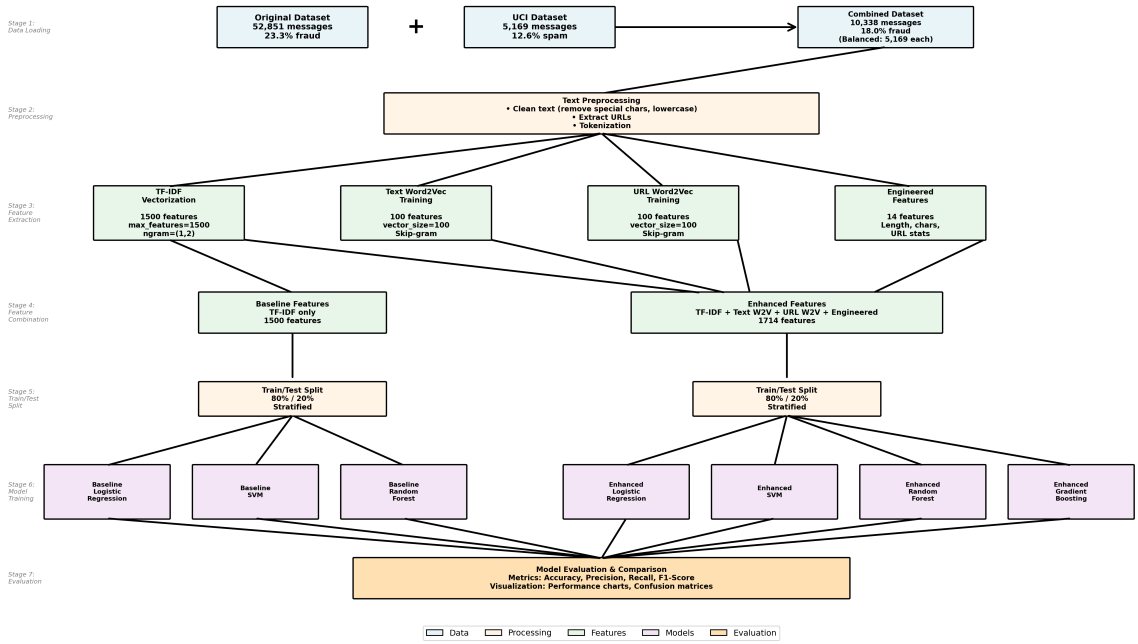

Figure 8 demonstrates an overall illustration of the architecture, demonstrating how raw SMS messages flow through preprocessing, feature extraction, and into machine learning models.

Having defined our features, we now describe how they're combined and used for training.

Feature Combination:

- TF-IDF: 1,500 features (keyword importance)
- Text Word2Vec: 100 features (semantic relationships in text)
- URL Word2Vec: 100 features (malicious domain patterns)
- Engineered Features: 14 features (fraud-specific indicators)
- **Total:** 1,714 features

Models Evaluated:

- **Baseline:** TF-IDF only (1,500 features)
  - Logistic Regression, SVM, Random Forest, Gradient Boosting
- **Enhanced:** TF-IDF + Dual Word2Vec + 14 Features (1,714 features)
  - Same four models for fair comparison

This modular architecture allows us to compare baseline (TF-IDF only) against enhanced (all features) approaches.

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3. Results

3.1 Overall Performance Comparison

Our enhanced approach achieves significant improvements over baseline methods. Enhanced SVM emerges as the clear winner with 99.37% accuracy and 97.04% recall.

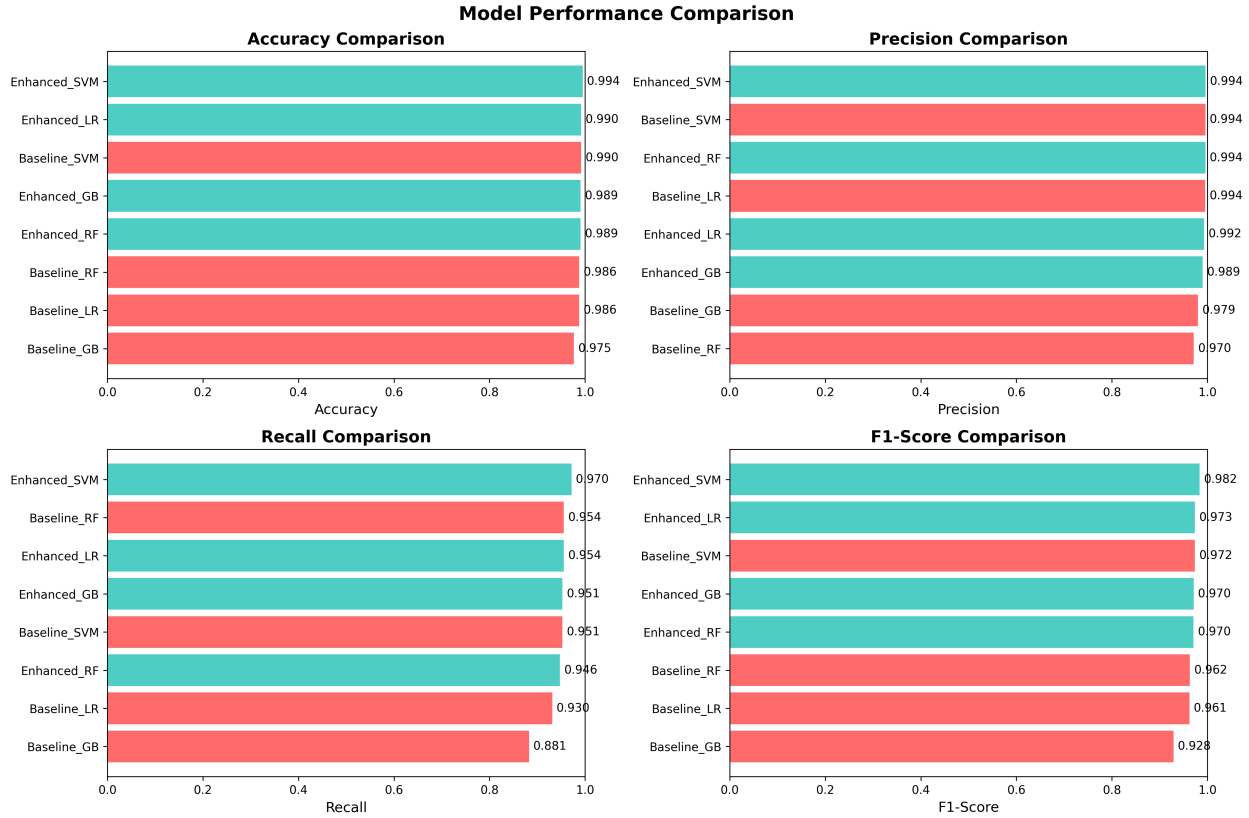

Figure 9 shows the comparison of training results of models across all eight configurations (four baseline, four enhanced).

The chart clearly shows that enhanced models (blue bars) consistently outperform their baseline counterparts (orange bars) across all metrics. Enhanced SVM achieves the best balance of accuracy, precision, and recall.

Key Observations:

1. Enhanced SVM is the clear winner
   - Highest accuracy: 99.37%
   - Highest recall: 97.04%
   - Best balance of precision and recall
2. All enhanced models outperform baselines
   - Average recall improvement: +2.1%
   - Consistent gains across all metrics

3.2 Feature Ablation Study: Proving the Value of Engineered Features

To validate that our 14 engineered features contribute meaningfully, we conducted an ablation study comparing models with and without these features.

Experimental Setup:

- Without Engineered Features: TF-IDF (1,500) + Dual Word2Vec (200) = 1,700 features
- With Engineered Features: TF-IDF (1,500) + Dual Word2Vec (200) + Engineered (14) = 1,714 features

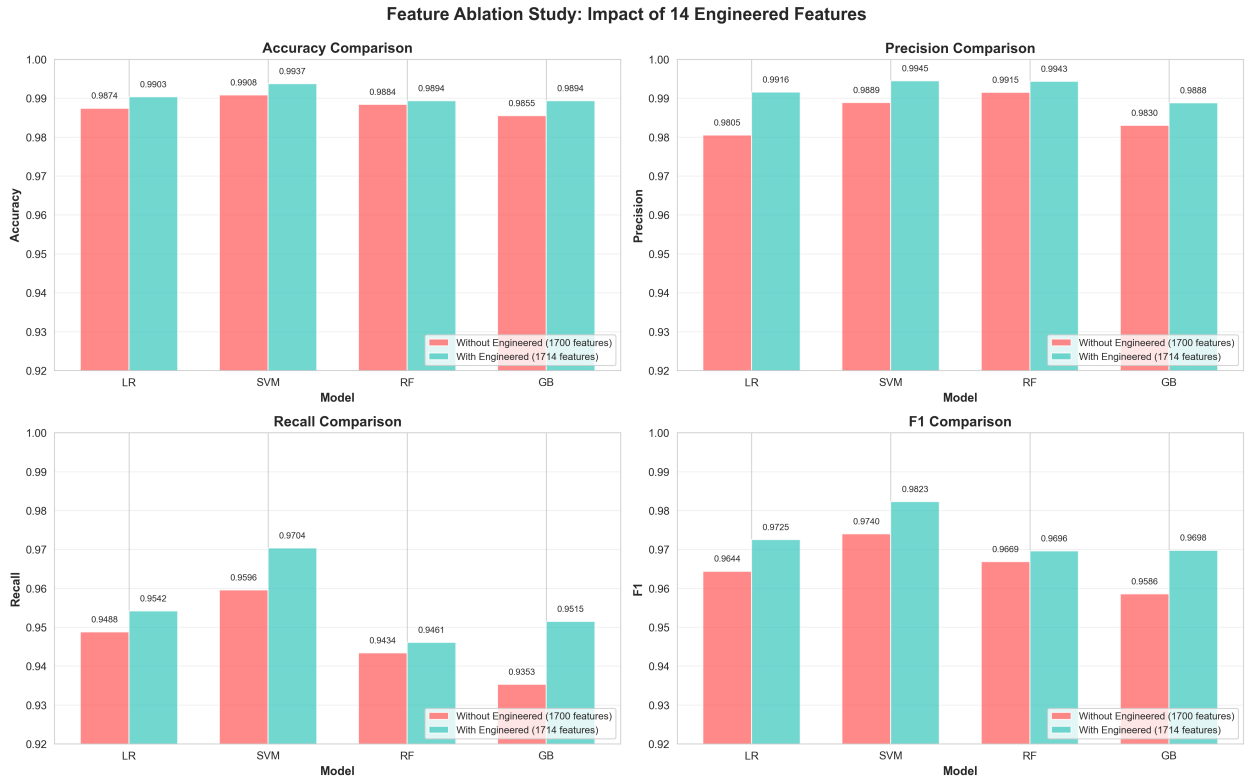

Figure 10 compares model performance with and without the 14 engineered features, demonstrating their impact on recall.

The 14 engineered features provide disproportionate value by capturing fraud patterns that semantic models miss.

Key Findings:

Just 14 carefully designed features (0.8% of total) improved recall by 1.12%, meaning 10 additional fraud cases caught per 10,000 messages.

Model	Metric	Without (1,700)	With (1,714)	Improvement
SVM	Accuracy	99.08%	99.37%	+0.29%
SVM	Recall	95.96%	97.04%	**+1.12%**

3.3 Model Selection: Enhanced SVM Wins

Enhanced SVM emerges as the optimal choice for production deployment due to several key advantages. It achieves the best overall performance across all metrics—highest accuracy, recall, precision, and F1-score—demonstrating superior fraud detection capability. The model is robust to high-dimensional data, effectively handling our 1,714-feature space without overfitting. Its kernel trick enables it to capture non-linear patterns in the data, making it particularly effective with imbalanced datasets where fraud cases are rare. Finally, SVM offers fast inference times suitable for real-time detection in production environments, processing messages with minimal latency.

3.4 Recall Matters In fraud detection, false negatives (missing fraud detection) are far more costly than false positives (legit cases incorrectly flagged as fraud):

Quantified Benefits:

Assuming 1 million SMS messages processed daily:

- Fraud rate: 1% (10,000 fraud messages)
- Baseline recall: 95.15% → 485 fraud messages missed
- Enhanced recall: 97.04% → 296 fraud messages missed
- Additional fraud caught: 189 messages/day

Annual Impact:

- 189 messages/day × 365 days = 68,985 additional fraud cases prevented
- Average fraud loss: $500
- Total savings: $34.5 million/year

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4. Conclusion

This project successfully extended the SMS fraud detection methodology of Abo Yehya and Salhab (2023) to address modern fraud patterns. We created a balanced dataset with a 50:50 split of URL-based and social engineering fraud, addressing the original dataset's 94.76% URL bias. Our enhanced SVM model achieved 99.37% accuracy and 97.04% recall—a 1.89% recall improvement over baseline—through dual Word2Vec models capturing semantic relationships, 14 engineered features providing 1.12% recall improvement, and POS-aware preprocessing preserving contextual meaning. Feature ablation studies validated our approach across multiple metrics using reproducible methodology with public datasets. Key insights for practitioners include: TF-IDF alone is insufficient for modern fraud detection; Word2Vec captures semantic relationships that keyword matching misses; engineered features encode domain-specific fraud patterns; recall should be prioritized over precision in fraud detection; and training on diverse fraud types improves generalization. Current limitations include the static model requiring retraining for new fraud patterns, language-specificity to English messages, and computational cost of 1,714 features requiring significant memory.

5. Reproducibility

All code, datasets, and documentation are publicly available:

Repository: https://github.com/dongk7york/SMS-Fraud-Detection-Enhanced

Key Files: - `sms_fraud_enhanced.py`: Main training script - `feature_ablation_study.py`: Ablation study implementation - `baseline_vs_enhanced_results.csv`: Complete performance metrics - `charts/`: All visualization outputs - `CITATIONS.md`: Detailed references and attributions

Requirements:

```
pip install -r requirements.txt
python sms_fraud_enhanced.py
```

References

Primary Sources

1. **Abo Yehya, B., & Salhab, N.** (2023). Telecommunications Fraud Machine Learning-based Detection. *2023 4th International Conference on Data Analytics for Business and Industry (ICDABI).*

Datasets

2. **UCI SMS Spam Collection Dataset**
   - URL: https://archive.ics.uci.edu/ml/datasets/SMS+Spam+Collection
   - DOI: https://doi.org/10.24432/C5CC84
   - License: Public Domain
3. **Original Telecom Fraud Dataset**

- Source: https://github.com/nsalhab/TelecomFraudDetection
- Author: Nour Salhab

Methodological References

4. **Mikolov, T., Chen, K., Corrado, G., & Dean, J.** (2013). Efficient Estimation of Word Representations in Vector Space. *arXiv preprint arXiv:1301.3781*.

Industry Reports

5. **Communications Fraud Control Association (CFCA)**. (2021). CFCA 2021 Fraud Loss Survey. Global telecommunications fraud losses: $39.89 billion.

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Contributions

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